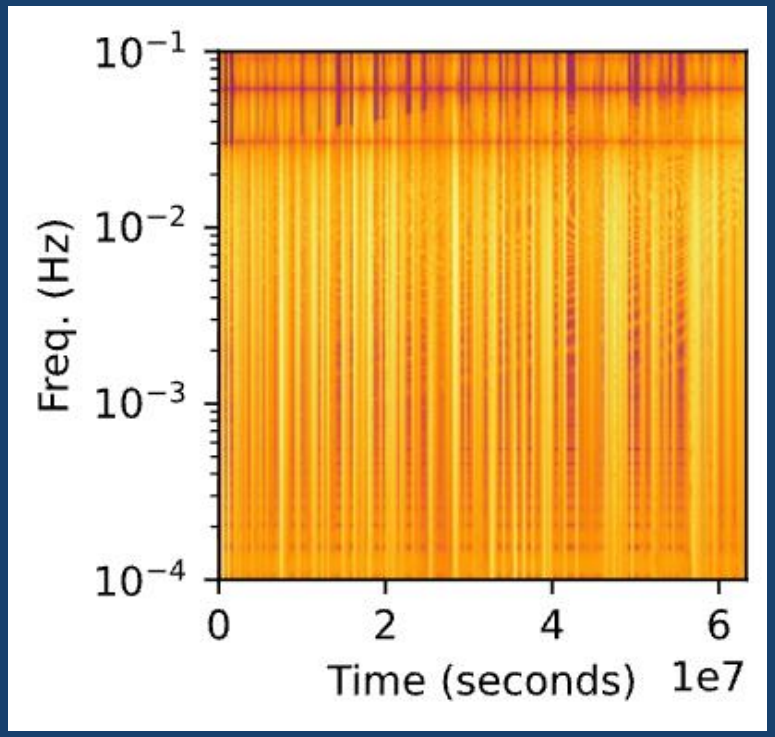
Noisy GW signal

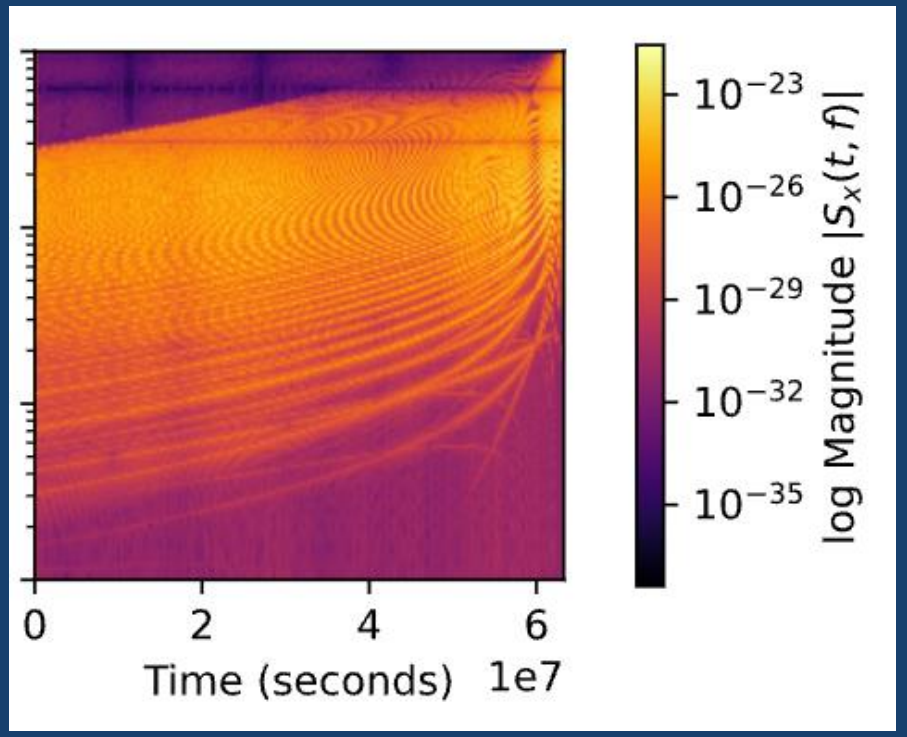


Data cleaner

Input

Output

Clean GW signal



Wavelet-domain Masking of Instrumental Noise Artefacts from Extreme Mass Ratio Inspirals

Amin Boumerdassi

NZ Gravity Workshop, Auckland | 06/11/26

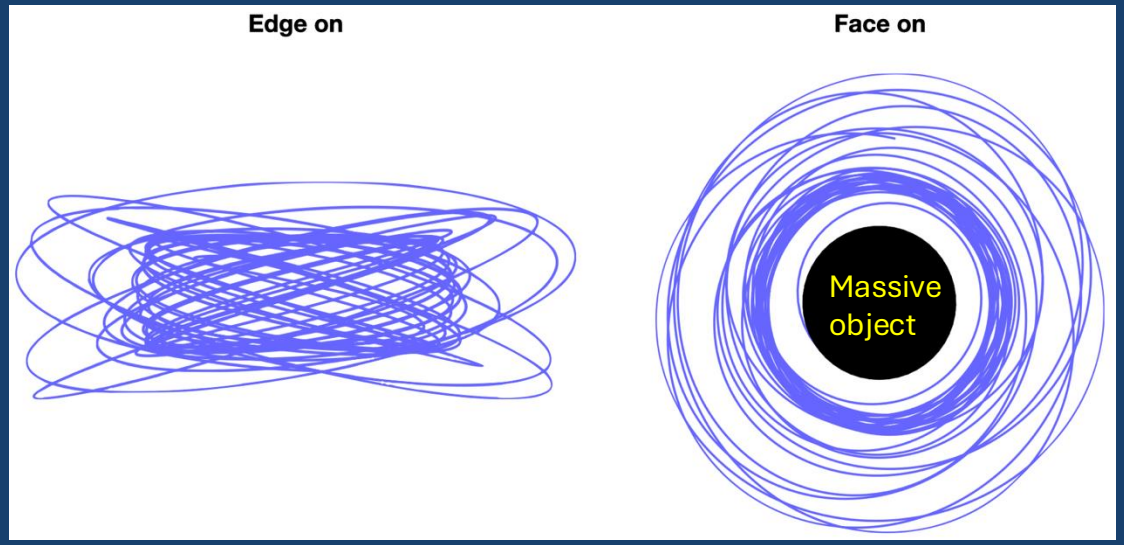
LISA CONSORTIUM

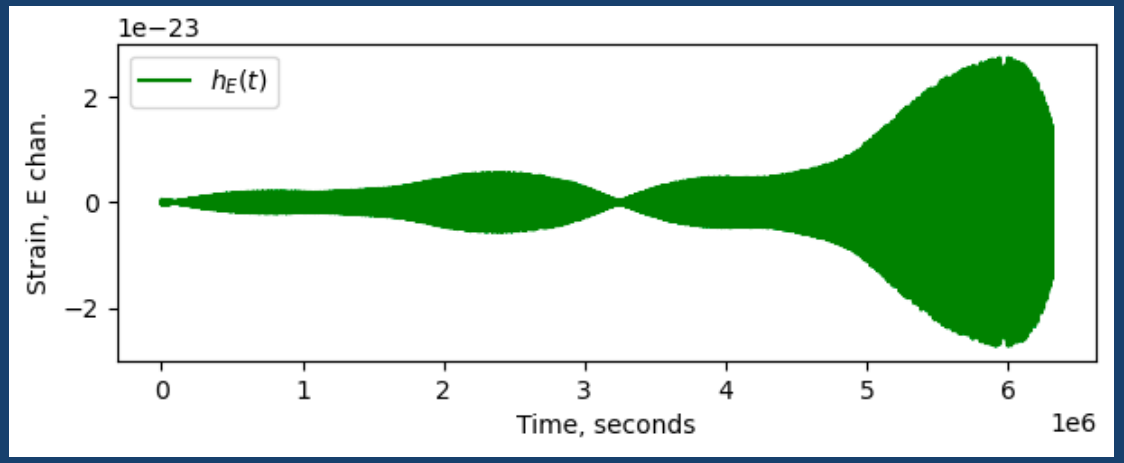
THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

GW source: extreme mass ratio inspiral

- Slow inspiral of massive and compact object with mass ratio $\frac{\mu}{\mathrm{M}} > 10^{-4}$
- Low-amplitude, years-long GW signals
- Successful detection will permit new tests of GR, cosmology etc.

EMRI's $h(t;\bar{\theta})$ is modelled with 14 parameters e.g. masses M and μ, spin a, inclination ι, ...



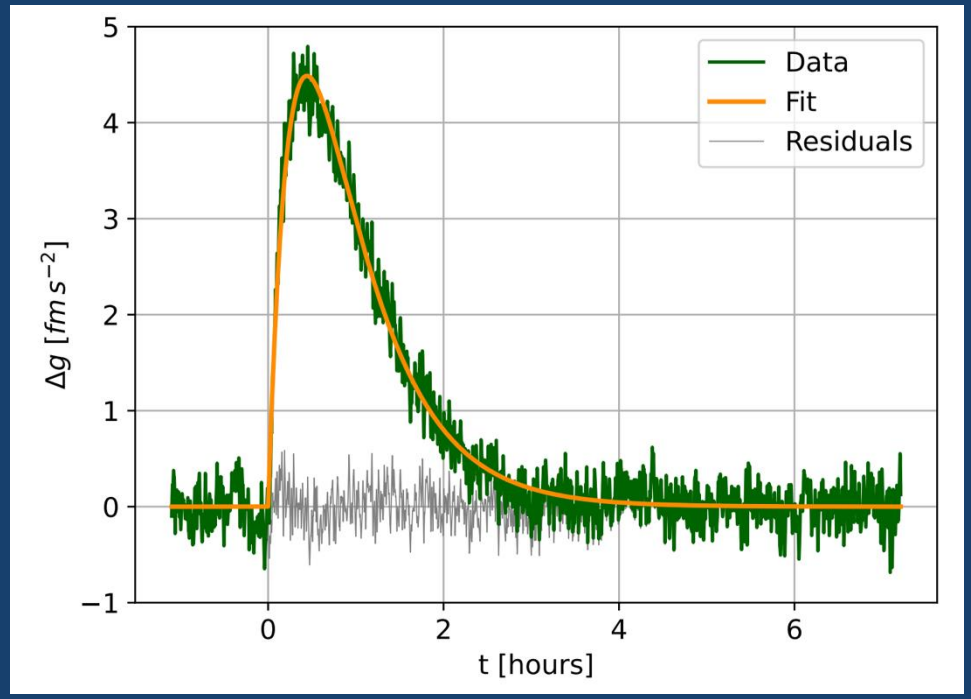


GW noise: glitches

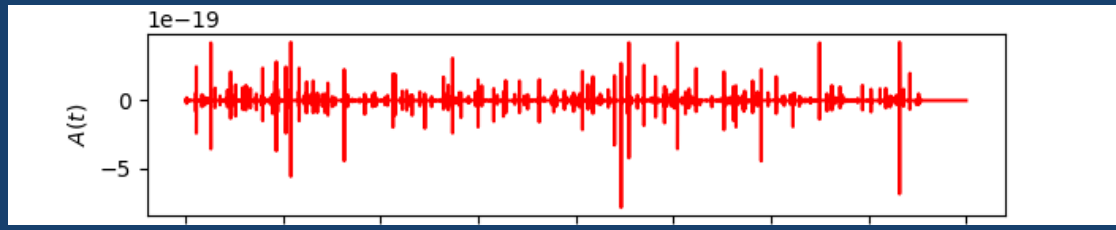
- Short, loud instrumental noise artefacts
- NOT astrophysical
- Generally unexplained or coming from local disturbances to the detector

Risks that glitches pose:

- obscure or partially obscure genuine GW events
- mistaken for genuine GW events
- induce biases in GW parameter estimation



Fast Rising Exponential Decay (FRED) glitch



Goals & aims

Successfully mitigate glitches from noisy EMRI data such that EMRI parameter estimation becomes:

- Minimally biased
- Maximally precise

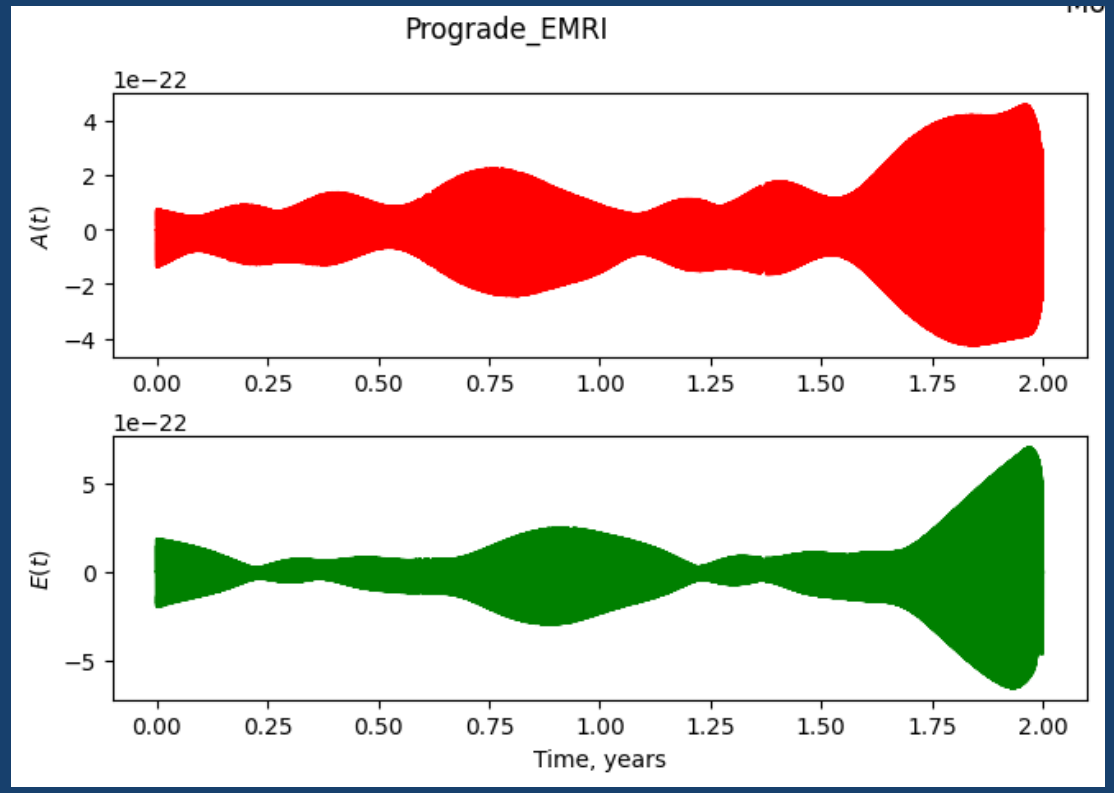
For this:

- Develop a glitch mitigation algorithm that removes *ensembles* of glitches

Modelling EMRIs

Dataset of 10,000 EMRIs:

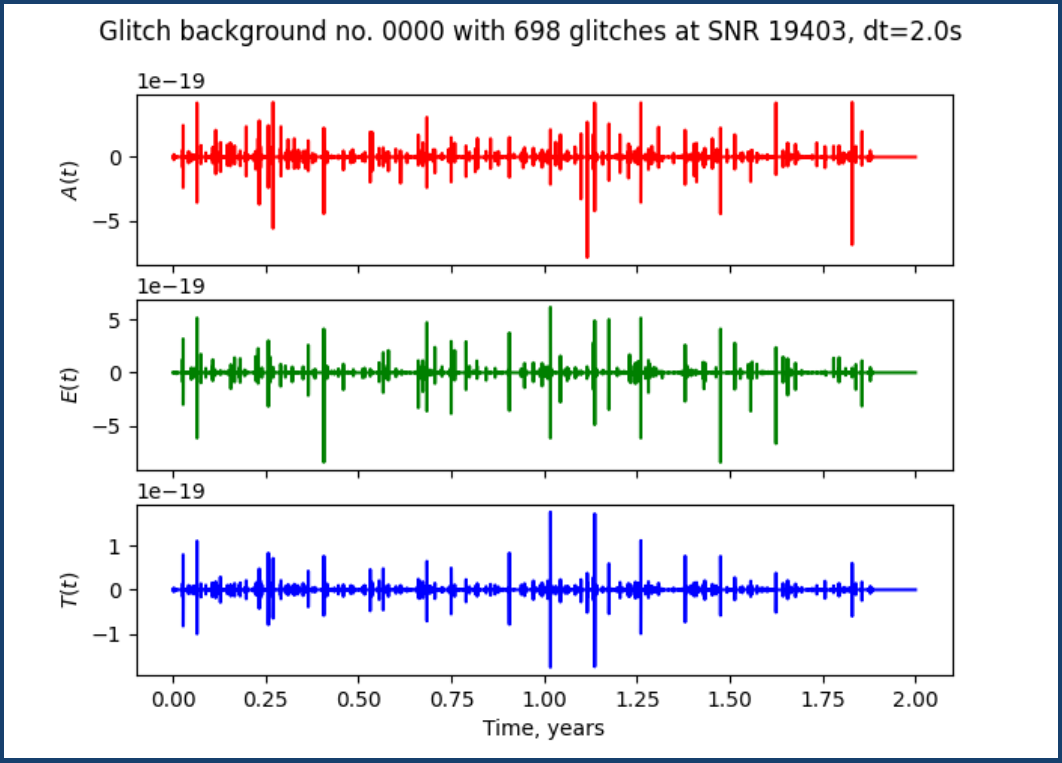
- "FastSchwarzchildEccentricFlux" model
- Mass ratios $\frac{\mu}{\mathrm{M}} \sim U[10^{-5}, 10^{-4}]$
- Non-spinning EMRIs
- Initial eccentricities $\sim U[0,0.7]$
- No inclination
- Optimal SNRs $\sim U[20,80]$



Modelling glitch backgrounds

Dataset of 128 glitch backgrounds:

- $N_{glitches} \sim \mathrm{Po}(\lambda = 1 \text{ per day})$
- Arrival time ~ Exp($\lambda = 1$ per day)
- Amplitude and duration resampled from empirical distribution (LISA Pathfinder)



Modelling the data (in time-frequency)

The data is a sum of:

1. The true EMRI $h(t, f; \theta_0)$ parameterised by true parameters θ_0.
2. Glitch background $g(t, f)$ i.e. an ensemble of glitches with variable count, arrival times, amplitudes, durations etc.

Our data is:

$$d(t, f) = h(t, f; \theta_0) + g(t, f)$$

- Apply some kind of time-freq. transformation e.g. STFT, wavelet etc.

Denoising through mask estimation

Borrowing from audio separation works:

1. Work in the time-frequency domain
2. Adopt a multiplicative “denoising mask” (various types!)
3. Leverage neural networks to learn these masks from the data

Define a mask $M(t,f)$ such that

$$M(t,f) \odot d(t,f) \approx h(t,f) \therefore M(t,f) = \frac{h(t,f)}{d(t,f)}$$

$$M \coloneqq \text{complex ideal ratio mask}$$

Denoising through mask estimation

The neural network acts a function H that takes input d and outputs a predicted mask $\widehat{M}$.

$$H(d; \bar{w}_{NN}) = \widehat{M} \approx M$$

The loss function is:

$$L = \mathrm{MAE}(\widehat{M}, M) + \alpha(1 - \mathrm{Overlap}(\widehat{M} \odot d, h))$$

weighting

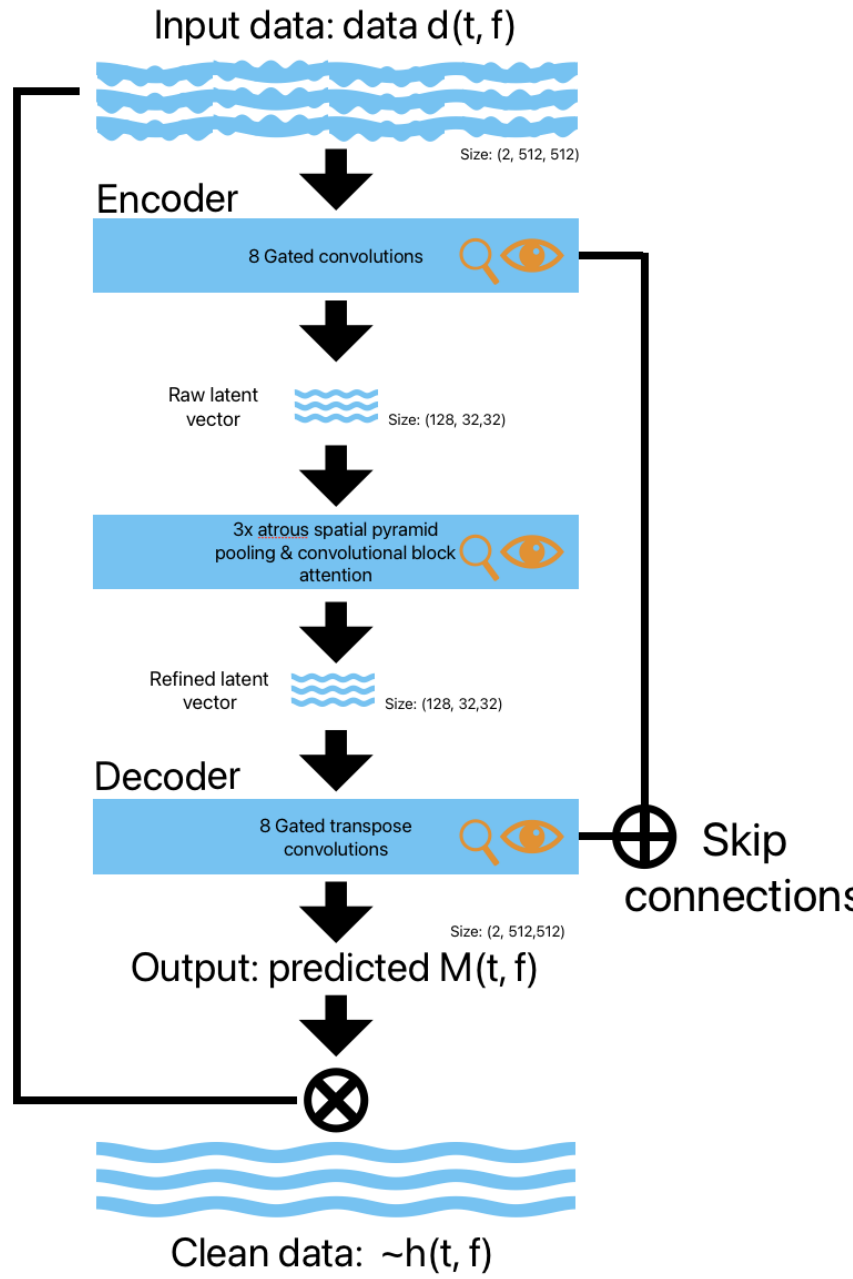
Point-wise accuracy

"correlative" accuracy

$$\mathrm{Overlap}(a, b) := \frac{(a|b)}{\sqrt{(a|a)(b|b)}} \qquad (a|b) := 4Re \int \frac{a(f)b^*(f)}{S_n(f)} df$$

Brief details on the neural net

- Purely convolutional U-net style model
- Components: encoder, bottleneck, decoder
- Decoder layers directly receive information from encoder layers
- Learn patterns in the data to extract the correct mask



Results

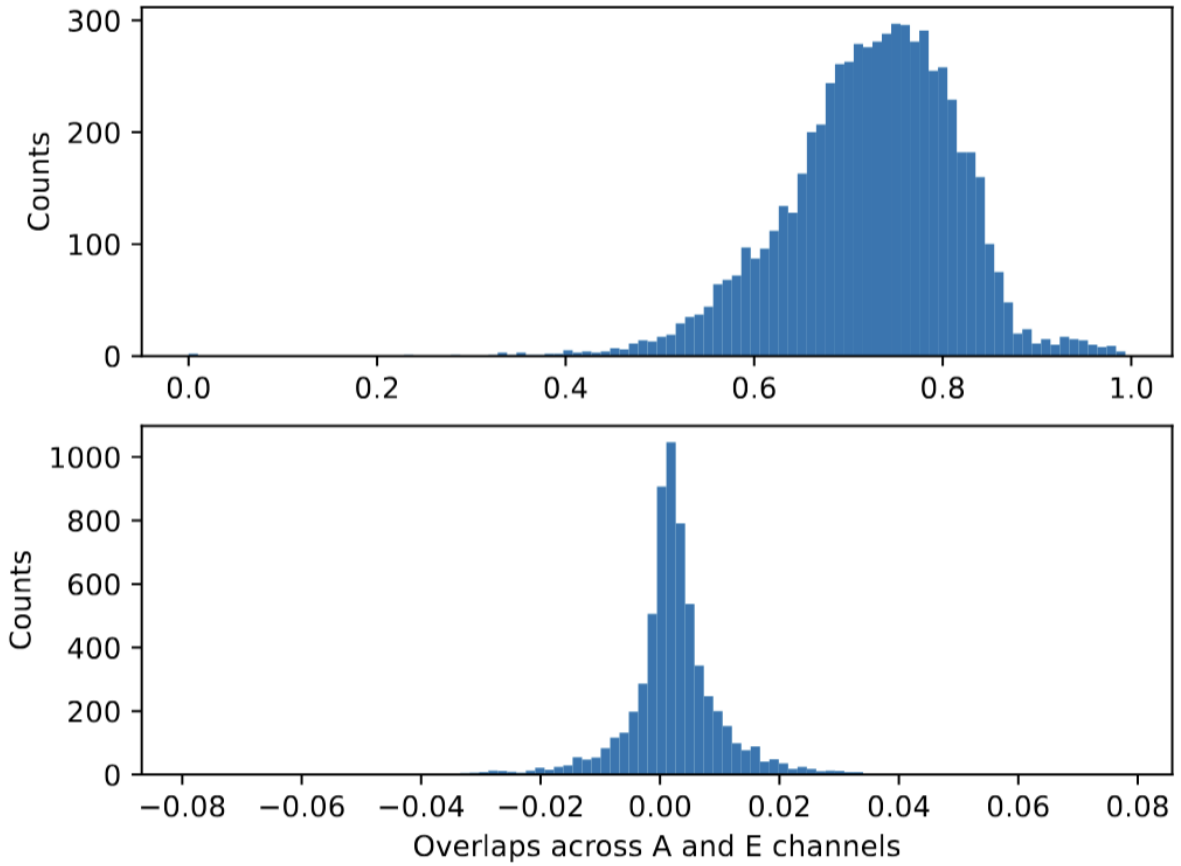


Figure 1.2: *Histograms of 3200 overlaps across the A and E channels for the denoised data (top) i.e.* $\mathcal{O}(\widehat{M} \odot d, h)$ *and the original noisy data (bottom) i.e.* $\mathcal{O}(d, h)$.

After: Average overlap of ~0.7

Before: Average overlap of ~1e-3

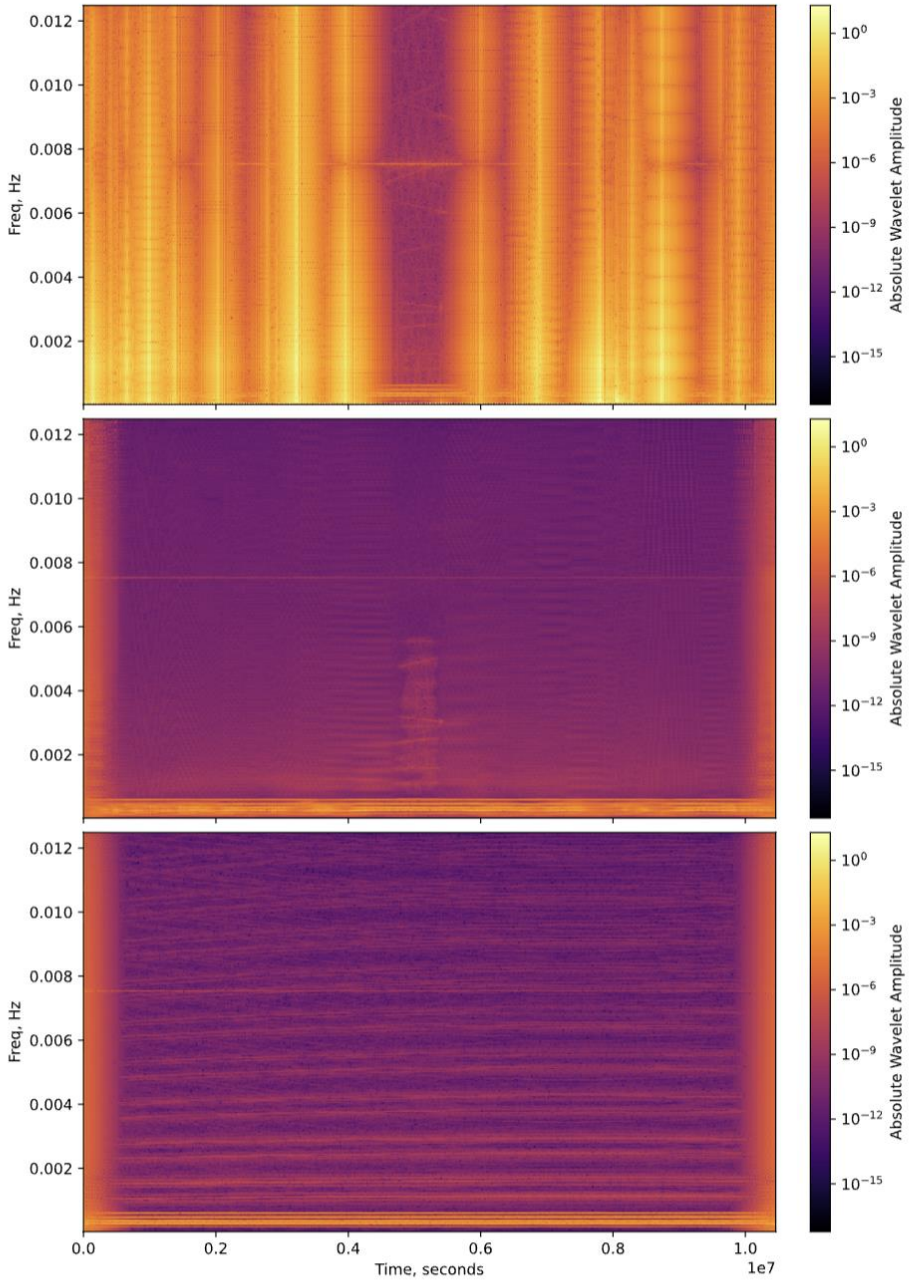


Figure 1.4: *Wavelet scalograms in the A channel of a noisy EMRI (top), the denoised EMRI (middle), and the clean EMRI (bottom). This denoised EMRI had an overlap of* 0.71.

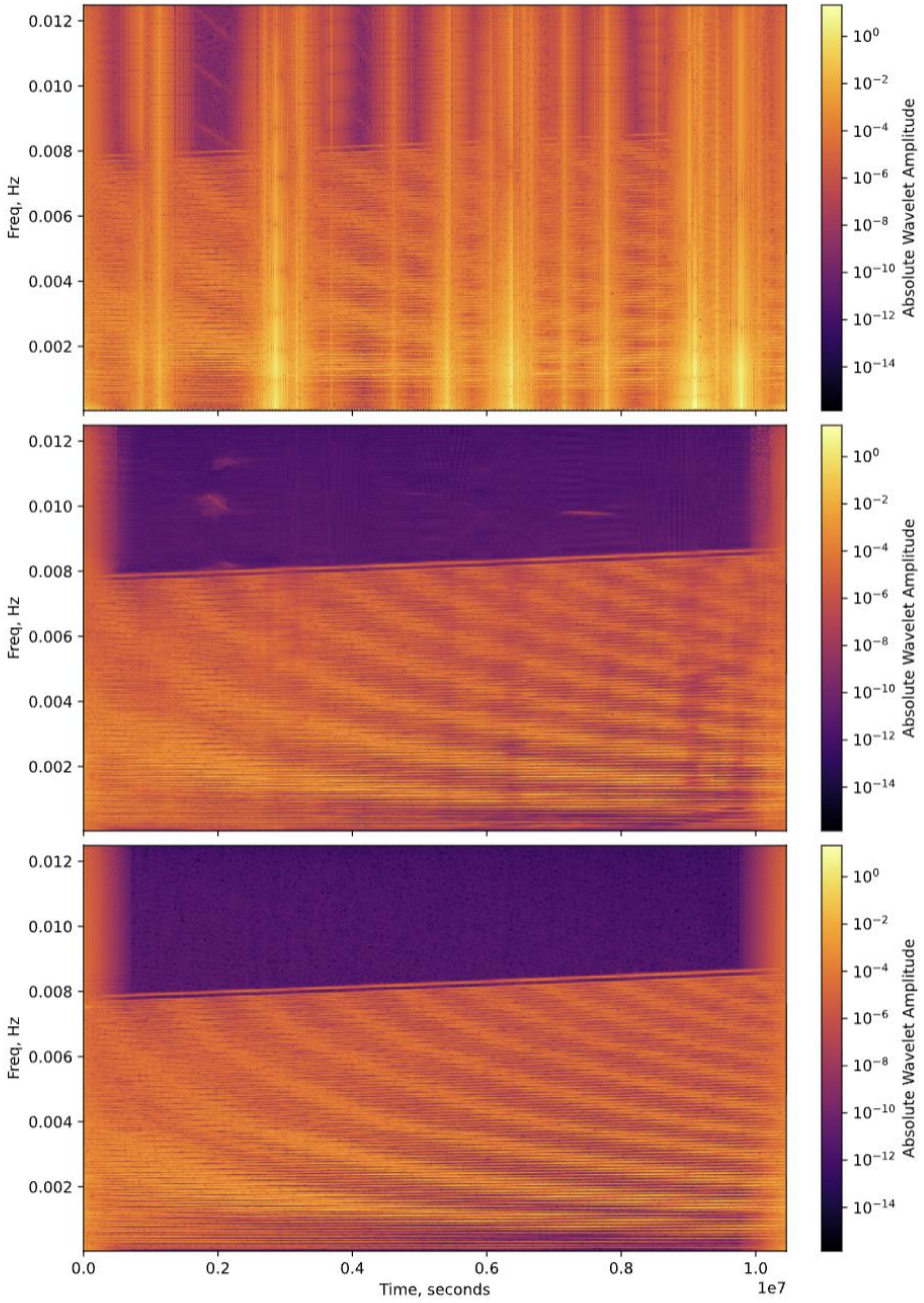


Figure 1.3: *Wavelet scalograms in the A channel of a noisy EMRI (top), the denoised EMRI (middle), and the clean EMRI (bottom). This denoised EMRI had a relatively high overlap of* 0.87.

Conclusions & next steps

- Wavelet-domain masking works on EMRIs in principle, though imperfectly
- Limited by method or model?
- Can the overlaps be pushed to the gold standard of >0.95?
- Does the masking reduce inference bias?

In future:

- Investigating alternative masking methods i.e. in the latent domain